

Data Quality-aware Mixed-precision Quantization via Hybrid Reinforcement Learning

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Abstract—Mixed-precision quantization mostly predetermines the model bit-width settings before actual training due to the non-differential bit-width sampling process, obtaining sub-optimal performance. Worse still, the conventional static quality-consistent training setting, i.e., all data is assumed to be of the same quality across training and inference, overlooks data quality changes in real-world applications which may lead to poor robustness of the quantized models. In this paper, we propose a novel Data Quality-aware Mixed-precision Quantization framework, dubbed DQM, to dynamically adapt quantization bit-widths to different data qualities. The adaption is based on a bit-width decision policy that can be learned jointly with the quantization training. Concretely, DQM is modeled as a hybrid reinforcement learning (RL) task that combines model-based policy optimization with supervised quantization training. By relaxing the discrete bit-width sampling to a continuous probability distribution that is encoded with few learnable parameters, DQM is differentiable and can be directly optimized end-to-end with a hybrid optimization target considering both task performance and quantization benefits. Trained on mixed-quality image datasets, DQM can implicitly select the most proper bit-width for each layer when facing uneven input qualities. Extensive experiments on various benchmark datasets and networks demonstrate the superiority of DQM against existing fixed/mixed-precision quantization methods.

Index Terms—Network Quantization, Reinforcement Learning, Model Compression, Bit-width Decision, Data Quality.

I. INTRODUCTION

Model quantization has attracted increasing attention recently as an effective solution that improves the computation and memory efficiency for deep neural networks (DNN), thus enabling the deployment of deep models on resource-constrained edge devices [2, 3, 4, 15, 19, 37, 38, 39]. Specifically, the lower description bit-width of DNN can bring about significant benefits in terms of inference acceleration and model compression rate, for example, Int8/Int4 quantization and binary neural networks [6]. However, it inevitably causes a greater accuracy drop than higher description bit-width models due to unavoidably more information loss.

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To maximally excavate the description redundancy and learn the compressed model for a given training domain, mixed-precision quantization [2, 3, 7, 8, 9, 10, 20] has been proposed to explore a better balance between task performance and model description length. It allows network components, e.g., layers [14] or kernels [20], to be quantized with lower bit-widths instead of full precision bit-widths if they have less contribution to model accuracy, wherein, more sensitive components are quantized with relatively higher bit-widths to avoid large accuracy degradation and vice versa.

Despite the more balanced compression power, the mixed bit-widths can bring about unique challenges for the quantization setting exploration and model training. Concretely, the search space for determining a sufficiently good mixed-precision quantization setting can grow exponentially with increasing number of layers and bit-width choices. For example, when a ResNet-152 is quantized layer-wisely with 4 possible bit-widths, the search space can easily reach 4^{152} times. Worse still, the inherently discrete and non-differentiable bit-width decision-making (or sampling process) makes it hard to learn the bit-width policy jointly with quantization training. To deal with the above challenges, existing state-of-the-art (SOTA) works usually decouple the non-derivable bit-width sampling process from the end-to-end supervised training, i.e., consisting of two mutually independent precision decision-making stage and quantization training stage. Some representative methods include state-of-the-art HAWQ series [3, 7, 8], DMBQ [9], FracBits [10], AutoQ [20], etc.

For example, The HAWQ series [3, 7, 8] measure the relative sensitivity of layers via layer-wise hessian information and develop a Pareto frontier based method to make hessian-based bit-width decisions before model quantization. Differently, DMBQ [9] tries to search for an optimal bit-width setting over the distribution space beforehand and quickly selects one setting via this prepared fast lookup table during training. In general, previous These state-of-the-arts make satisfying bit-width decisions under the guidance of hardware feedback [14] or gradient-based information [7] before the actual quantization training, as well as the practical inference, while all their bit-width selection and quantization training are done in the given narrow data domain with quality-consistent training setting.

In real-world application, such decoupled training process can inevitably lead to the data isolation issue between three stages of precision decision-making, quantization training, and practical inference, which can result in sub-optimal performance. Besides, the static quality-consistent training setting is